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Krishna Balaji S

Education

2024–2026 **M.Tech. Quantum Science and Technology**, *Indian Institute of Technology Madras*, CGPA: 8.78/10 (dual degree); Master's thesis: 10/10

2021–2025 **B.Tech. Engineering Physics**, *Indian Institute of Technology Madras*

Research Experience

Nov 2025–Feb 2027 **Full-time Researcher**, *Max Planck Institute of Quantum Optics*, Garching
Guide: Prof. Monika Aidelsburger

Aug 2026–Feb 2027 *A. Predoctoral Scholar: Yb double-frequency clock experiment*

- Built and characterized an AOM board with double-pass setups for the Zeeman slower, 2D-MOT and imaging beams at 399 nm.

Nov 2025–May 2026 *B. Master's internship: Designing a high-flux source of Yb atoms*

- Contributed to the design of a vacuum chamber and simulated the atomic physics to numerically optimize the magnetic field configuration, Zeeman slower and 2D-MOT parameters for a high-flux Yb atomic beam source.
- Characterized current coils, temperature sensors and permanent magnets; specified and procured AOMs, amplifiers, optics and optomechanics for the cooling and trapping setup.

Mar 2025–Aug 2026 **Master Student/Collaborator**, *Quantum Optics Theory Group, IIT Madras*
Guide: Prof. Athreya Shankar

Jun 2026– *A. GPU-accelerated variational dynamics for open many-body systems*

- Implemented a GPU-based TDVP solver, completing a full three-dimensional laser-cooling simulation in 20 minutes on an NVIDIA RTX 5000 Ada.
- Reproduced exact master-equation results for the driven-dissipative Dicke model, demonstrating multi-spin and multi-mode capability of the ansatz.

Jun–Aug 2026 *B. EIT cooling dynamics at high motional occupation (manuscript in preparation)*

- Extended the grid-based formalism towards efficiently simulating EIT cooling dynamics of a single ion under harmonic confinement.
- Extracted EIT cooling rates up to $\max(\bar{n}) \gtrsim 2000$, a regime inaccessible to most exact solvers.

Mar 2025–Jun 2026 *C. Master's thesis: Quantum dynamics of neutral atoms in optical dipole traps*

- Developed a solver using the sum-of-Gaussians time-dependent variational principle (TDVP) ansatz and applied it to two-dimensional laser-cooling dynamics, running over 4× faster than QuTiP's `mcsolve` at matched trajectory count.
- Developed a grid-based quantum Monte Carlo solver for dissipative dynamics of single-mode multi-level systems, and studied state-dependent trapping to identify regimes where the ^{171}Yb qubit can benefit from imperfect trapping.
- Benchmarked against the experimental work of Muzi Falconi et al. for ^{171}Yb atoms in an optical tweezer, accurately identifying the non-thermal dynamics of energy distribution in the trap. *Co-supervised with Prof. Monika Aidelsburger (MPQ).*

May–Aug 2024 **USEQIP Fellow**, *Institute for Quantum Computing*, University of Waterloo
Guide: Prof. Alexandre Cooper-Roy, Rydberg Atom Quantum Simulator Lab

Site-Selective Optical Control in a ^{87}Rb Quantum Simulator

- Built a 795 nm optical system for site-selective single-qubit gating in a ^{87}Rb quantum simulator.
- Implemented and characterized an optical phase-locked loop between the qubit rotation lasers at the 6.83 GHz qubit frequency, measuring the residual phase-noise spectrum.
- Generated a 1D tweezer array by optimizing AOD diffraction efficiency and aligning it to the objective.

Aug 2023–Oct 2024 **Young Research Fellow**, *Femtosecond Laser Facility*, IIT Madras

Guides: Prof. Sivarama Krishnan and Prof. Vidya Praveen Bhallamudi

Single-Photon LiDAR and Computational Imaging

- Built a single-photon LiDAR system with SPAD and TCSPC electronics, and a BBO second-harmonic stage converting 1064 nm to 532 nm.
- Performed a literature review of reconstruction algorithms and achieved sub-cm resolution by modelling the SPAD instrument response as an exponentially-modified Gaussian (EMG).

Publications

In prep. Manuscript in preparation on EIT cooling dynamics of trapped ions, with A. Shankar (IIT Madras) and J. Zaris (University of Colorado Boulder).

Honors, Awards and Schools

Jul 2026 **Introductory Course on Ultracold Quantum Gases**, *University of Innsbruck*

May 2026 **Workshop on Kähler Geometry**, *Solvay Institutes, Brussels*

2025 **ISIT Quantum Hackathon**, *IEEE International Symposium on Information Theory*

Third place, for a QAOA and quantum-annealing approach to intelligent reflecting-surface beamforming.

2024 **USEQIP Fellowship**, *Institute for Quantum Computing, University of Waterloo*

2023–2024 **Young Research Fellow**, *Indian Institute of Technology Madras*

Selected as one of 28 Young Research Fellows across all undergraduates in the 2023–24 cohort.

2023 **Mimamsa**, *Third place, India's largest science quiz competition*

2021 **JEE Advanced**, *All India Rank 833 among 150,000 candidates*

Leadership and Outreach

Sep 2024–Sep 2025 **Founder, Quantum Computing Society (QCSoc)**, *IIT Madras*

- Founded and led a society that drew over 700 students across disciplines to quantum computing, running sessions on architectures, algorithms and error correction.
- Presented an intuitive route into superconducting and neutral-atom architectures, from magnetic-resonance ideas and school-level prerequisites, to audiences nationwide.

Jan–May 2025 **Teaching Assistant, PH5500 Dynamical Systems**, *IIT Madras*, Prof. Arul Lakshminarayan

Prepared solutions and evaluated examinations on discrete and continuous-time stability, chaos, and Hamiltonian dynamics for an advanced undergraduate course.

Mentoring and Outreach

Delivered a session on data-driven dynamics for nonlinear systems at the institute technical fest; personal mentor to four first-year students.

Technical Skills

Numerics: QuTiP, NumPy/SciPy, Numba, joblib, CuPy and PyTorch (CUDA), SLURM/HPC clusters

Methods: qMCWF for evolving Lindblad master equations, time-dependent variational principle (TDVP), convex optimization

Coursework:

Advanced QM	(S)	Quantum Error Correction	(A)
Atomic Physics	(S)	Quantum Experimental Techniques	(S)
Quantum Optics	(S)	Advanced Linear Algebra	(S)
Condensed Matter	(S)	Differential Geometry	(B)
Quantum Many-Body Physics*			

**Currently auditing at TUM. IITM grades follow a 10-point scale: S = 10, A = 9, B = 8.*

Laboratory: Laser locking and OPLL, AOM/AOD alignment, free-space optical design, magnetic field simulation and coil characterization, RF electronics

Languages: Python, C++, MATLAB, Mathematica